

CSE 400: Fundamentals of Probability in Computing
Project Scribe Submission

Group Number: S1_G7 ITS

Date: February 4th, 2026

Scribe Question 4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with the project’s implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

Answer: We use SUMO (Simulation of Urban MObility) to implement out probabilistic model. SUMO generates traffic flow and demand that creates randomness in the traffic which our model tries to control. “Modified Webster” formula is used to calculate an ideal cycle time for the traffic signals and then the Las Vegas randomized algorithm is incorporated to randomly select signal phases under the traffic conditions that were randomly generated by SUMO. Each phase that’s selected is verified deterministically for safety, timing constraints and clearance before being accepted.

Here we assume that traffic can be oversaturated, i.e., the demand to pass by can be more than the capacity of the traffic junction which in turn breaks the probabilistic queueing models; for this very reason, we implement the Las Vegas verification layer to reject phases that are infeasible and retries until a valid configuration is generated.

Scribe Question 5: Cross-Milestone Consistency and Change

Describe the current state of your project’s probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones.

Answer: The baseline probabilistic model is well defined, although the relationship between the intersections dependencies is currently undefined. As of now, we are only modelling an isolated intersection by assuming any dependency from the neighbouring intersection/signals. In the future milestones, we aim to refine the model to consider dependencies from other intersections such as the output of one intersection becomes the input for the other. This might require us to shift from a single isolated Las Vegas implemented signal to multiple signal in order for them to work in a network.

Scribe Question 6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project

at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Answer: The model will not fully account for the non-stationary arrival rates, like on-road accidents that lead to the model changing the arrival distribution parameters. In this situation, the Las Vegas randomized model only reacts to the current state but cannot predict the distribution shifts.

The biggest ambiguity this model faces is to accurately measure the loss time as a random variable instead of a fixed constant. This is due to its variability which directly affects the acceptance rate of the phases and the feasibility verification.

The Model Design Lead will be responsible for researching and developing way to integrate a prediction standard to anticipate trends in the distribution with guaranteeing zero-error, while the Simulation Lead must implement the traffic patterns in SUMO to test the model's limit.

End of Submission